SMITH ENGINEERING

Queen's University

Department of Electrical and Computer Engineering

ELEC 490/498 Project Proposal

Autonomous Smart Caddy Cart (ASCC)

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Submitted by Group 07

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Executive Summary

Walking golfers can already buy electric push carts that follow them around the course, but these carts only carry the bag. To know the distance to the green, or to keep track of their clubs, golfers still need a watch, a phone, or a separate tracker. The Autonomous Smart Caddy Cart (ASCC) puts both jobs on one cart, so the golfer can leave the phone in the bag.

The cart follows the golfer at a set distance, stops for obstacles, and can be stopped at any time with a remote emergency button. A reader in the bag knows which clubs are in their slots, so the cart can warn the golfer when one has been left behind. A display screen shows the distance to the green using GPS. None of this needs any input from the golfer during play. The project is deliberately limited to these three features. Extras such as shot tracking and club recommendations will only be attempted once they work.

The prototype will be judged against measurable targets. The main ones are holding a following distance of about 1.5 m, stopping within 200 ms of detecting an obstacle, and showing the distance to the green to within a few metres. The bag and the drive system will be tested separately before the whole cart is tested outdoors. The biggest technical risks are reading the club tags reliably inside the bag and losing track of the golfer. To manage them, the tag reader is tested early and the cart stops whenever it loses the golfer.

The deliverable is one working prototype, built on a used motorized golf cart and demonstrated on dry grass with one golfer and one bag of clubs. The cart should be driving under motor control by the end of October 2026, tracking clubs by December, and fully integrated and tested by March 2027. The estimated cost is 595 CAD, which fits the 600 CAD budget because the team already owns the Raspberry Pi and some of the sensors.

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Acronyms

ASCC Autonomous Smart Caddy Cart
GPIO general-purpose input/output
GPS Global Positioning System
I2C Inter-Integrated Circuit
SPI Serial Peripheral Interface
UART universal asynchronous receiver-transmitter

1. Introduction

1.1 The shortfalls of autonomous golf carts

Walking golfers can already buy electric push carts that follow or lead the way on the course [1], so the problem of carrying your own clubs has been solved. Where these carts fall short is that they can't tell you the distance to the green or notice a club left behind. Golfers get that information from a watch, phone, or range finder instead [2], [3], [4], usually with some manual input. The Autonomous Smart Caddy Cart (ASCC) is an all in one solution for the golfer who wants to keep track of their clubs and distances, stay off their phone, and not carry a separate tracking tool.

1.2 A cost effective, phone-free autonomous smart caddy

The objective is to design and prototype a golf caddy that follows the golfer safely, keeps an inventory of the clubs in the bag, and shows the distance to the green on a display screen, all without input from the golfer (Figure 1). Each of these features is measured against Table 3 and tested as described in Section 4: following distance (PF-03), obstacle stop (PF-01), remote stop (PF-05), club tracking (IF-03), and distance to the green (IF-02). The engineering involved is motor control, embedded sensing, golfer tracking, Global Positioning System (GPS) processing, and interface integration.

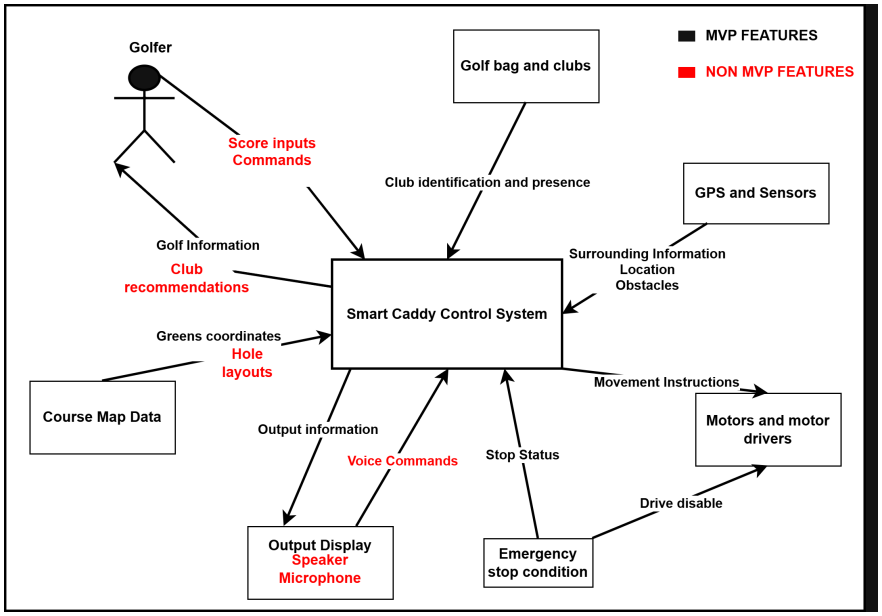


Figure 1: System context diagram for the Smart Caddy control system, showing the external entities it interacts with and the information exchanged with each.

Person-following robots track their target with cameras, laser rangefinders, or radio ranging [5], and commercial following carts track a handset that the golfer carries [1]. Hall et al. found that club-tag GPS trackers measure distances to within about three yards but can miss shots [4], so automatic shot logging is left as an extension. The ASCC instead detects clubs leaving the bag and shows distances on the cart. The trade-off is that it measures from the cart's position, not the golfer's. Table 1 summarizes the comparison, with details in Table 8 of Appendix A.

Table 1: Summary comparison of the proposed ASCC design with existing products.

Stewart Golf Q Follow [1]	Arccos Smart Sensors with Apple Watch [2], [3]	Proposed ASCC design
Following cart: carries the bag and follows the golfer, with no golf information	Golf tracker: tracks club use and GPS distances, with no bag transport	Both: follows the golfer, tracks the clubs, and shows the distance to the green

1.3 Deliverables: follow-me mobility, club inventory, and distance display

The deliverable is one working prototype cart, demonstrated on open, dry grass with one golfer and one bag of tagged clubs, a tagging strategy also used by commercial shot trackers [2]. Its inputs are the golfer's position, club tag readings, its GPS location, obstacle detections, and a remote stop button. Its outputs are wheel motion and a display showing the club inventory, missing-club warnings, and the distance to the green. The main limitations are dry-weather operation, restricted terrain, and distances measured from the cart, not the ball. Table 2 summarizes the deliverable, and Table 9 of Appendix A lists the assumptions, limitations, and uncommitted extensions.

Table 2: Summary of the committed deliverable.

Following	Safety	Club tracking	Distance
Powered cart follows one golfer on dry grass	Stops for obstacles and on a remote stop command	Detects each club's removal and return, and warns of a missing club	Shows the GPS distance to the green on the cart display

2. Functional Description

2.1 System overview

The system architecture is organized into functional, physical, and logical domains to validate core autonomous mobility, safety, and equipment auditing without unnecessary complexity. Figure 2 outlines the primary operational blocks, showing energy and signal paths between tracking, obstacle detection, club monitoring, and motor drive systems. Figure 3 defines the physical component distribution and interface boundaries across the chassis, emphasizing sensor clearance and immediate access to safety overrides (**IF-04**). Finally, Figure 4 shows how the software is layered between the hardware and the display, including the hardware communication protocols (UART, SPI, I2C, and GPIO), so that the cart stops safely when the emergency button is pressed (**PF-05**).

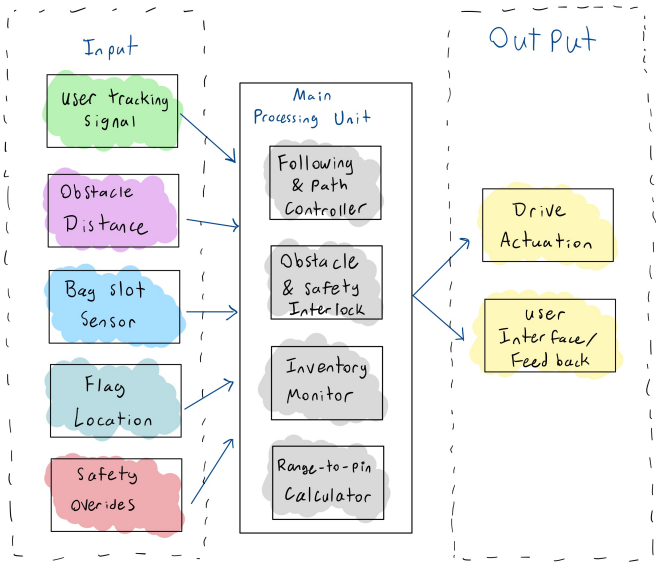


Figure 2: Functional architecture diagram showing inputs, processing interlocks, and control outputs for the MVP smart caddy system.

2.2 Interface and performance specifications

The interface and performance specifications in Table 3 set the measurable targets for the prototype. The interface specifications cover golfer tracking, distance accuracy, club sensing, the emergency-stop link, and the display (**IF-01** through **IF-05**). The performance specifications cover obstacle and emergency stopping, speed, following distance, and

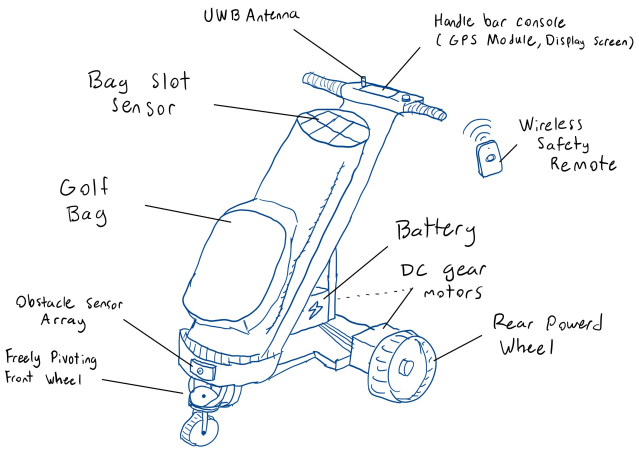


Figure 3: Physical layout diagram detailing component placement, structural enclosures, bag cradle sensors, and safety controls on the chassis.

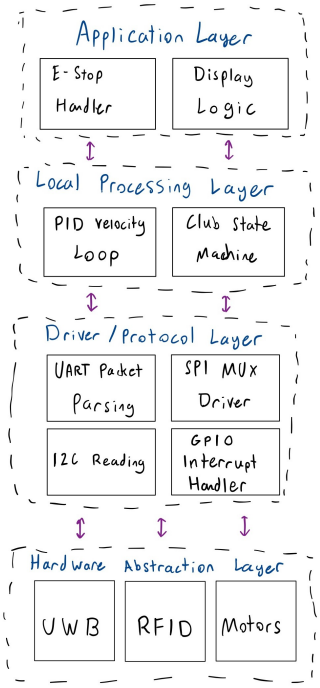


Figure 4: Logical computing architecture, from hardware abstraction to application layer.

range (**PF-01** through **PF-05**). Each specification has a unique identifier that the tests in Section 4 refer to.

Table 3: Preliminary system specifications.

ID	Requirement	Target	Tolerance
IF-01	User Tracking Beacon Range	15.0 m	$\pm$ 2.0 m
IF-02	Distance-to-Flag GPS Accuracy	3.0 m	$\leq$ 5.0 m
IF-03	Bag Slot Sensor Audit Resolution	14 slots	Binary (Present/Absent)
IF-04	Emergency Stop Wireless Link Range	30.0 m	Minimum 20.0 m
IF-05	Onboard Status Display Diagonal Size	7.0 in	Minimum 5.0 in
PF-01	Obstacle Auto-Stop Reaction Time	200 ms	$\leq$ 300 ms
PF-02	Maximum Autonomous Cruise Speed	5.0 km/h	$\pm$ 0.5 km/h
PF-03	Target Following Distance	1.5 m	$\pm$ 0.3 m

ID	Requirement	Target	Tolerance
PF-04	Continuous Operating Distance (Single Charge)	10.0 km	$\pm$ 1.5 km
PF-05	Emergency Stop Override Latency	50 ms	$\leq$ 100 ms

2.3 Risks and safety hazards

The main hazard of a self-driving cart is loss of control, where the cart moves outside its safe motion limits. Figure 6 in Appendix B maps the causes of this event, the preventative controls that make it less likely, and the protective controls that limit its consequences. Table 4 summarizes the remaining safety hazards and the main project risks, and Table 10 in the same appendix gives the consequence and proposed mitigation for each.

Table 4: Summary of safety hazards, project risks, and mitigations.

Safety hazards	Project risks	Mitigation approach
Collision, tipping on a slope, contact with moving parts, battery faults, and water reaching the electronics	Unreliable club-tag reads, insufficient GPS accuracy, and slow subsystem integration	Limited speed and stop behaviours, guarding and electrical protection, early subsystem tests, and optional extension features

3. Design and Production Approach

3.1 Development process

The autonomous golf pushcart will be developed through an incremental approach using system design, subsystem prototyping, integration, testing, and refinement. The team has already obtained some technical components (Raspberry Pi, some ultrasonic sensors, some motors) and will attempt to purchase a used motorized golf cart as the base platform. The Raspberry Pi, sensors, motor-control electronics, and smart-bag system will be integrated onto the cart, with development focused on motor control, sensor-based golfer following, RFID-based club identification and location. A controller will be used to process sensor inputs and control the cart's movement during golfer following and obstacle avoidance. Testing will begin in a controlled environment before progressing to outdoor conditions involving different walking speeds, obstacles, and terrain, with results used to refine the system and prioritize the MVP requirements.

3.2 Milestones and division of labor

Table 5: Technical milestones and ownership.

ID	Milestone	Due date	Lead
M1	Motorized control of the golf cart	Oct 30th, 2026	Connor
M2	Raspberry Pi integration and hardware integration of the necessary sensors	Nov 20th, 2026	Vince
M3	RFID-Based Club Identification and Tracking	Dec 18th, 2026	Jp
M4	Obstacle detection/avoidance/emergency stop functionality	Jan 22nd, 2027	Vince
M5	User Interface Design and display	Feb 5th, 2027	Jp
M6	Full Subsystem Integration and System Refinement	Mar 1, 2027	All members
M7	Testing and validation of the ASCC complete	Mar 15, 2027	All members

3.3 Contingencies

The main project risks include difficulty obtaining a suitable motorized cart, unreliable sensor-based golfer following, limitations in RFID detection, motor issues, and delays during system integration. If a component fails to meet requirements during testing, the team will modify the software or use an alternative sensor such as NFC tags, controller, or RFID configuration. If a suitable motorized golf cart cannot be obtained, a different cart or drive platform will be considered. Outdoor testing may be moved to a controlled environment if weather or terrain prevents reliable testing. If the project falls behind schedule, the team will prioritize the MVP features of basic cart movement, golfer following, obstacle detection, emergency stopping, and RFID-based club identification and tracking, while postponing more advanced features such as shot-based club recommendations or mobile-app functionality.

4. Testing and Evaluation

Testing is staged: each subsystem is tested on its own first, and the cart is tested as a whole once all of them work. Table 6 links each stage to the specification IDs in Table 3,

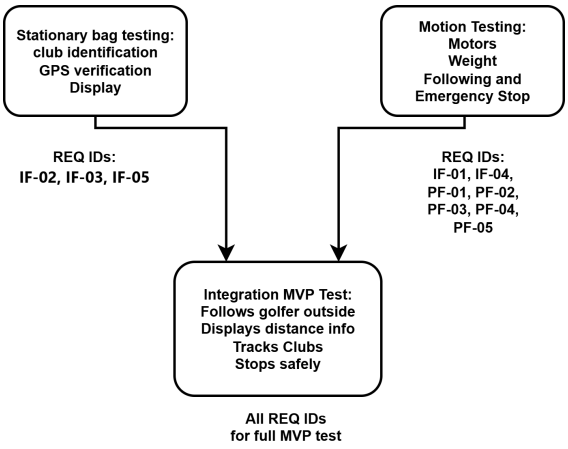


Figure 5: Proposed testing approach: stationary bag testing and motion testing are completed separately, with their requirement IDs, before the integrated MVP test.

and Table 11 in Appendix C gives the procedure and expected evidence of success for each. These tests verify the requirements, and they need only a controlled grass area, a tape measure, a laser rangefinder for reference distances, and video for timing. Validation is a final trial in which a golfer plays a hole using only the cart, with no phone. A safety operator holding the emergency stop is present for every test in which the cart moves.

Table 6: Summary of the proposed test stages and the specifications each one verifies.

Mobility and safety	Club identification	GPS and display	Range
IF-01, IF-04, PF-01, PF-02, PF-03, PF-05	IF-03	IF-02, IF-05	PF-04
Following, obstacle, and emergency-stop trials on grass	Club removal and return with the bag stationary	Displayed distance against reference distances	Loaded run on one charge

5. Resource Requirements

5.1 Scheduling access to resources

Purchased parts for the early milestones will be ordered as soon as possible. Outdoor testing needs no booking, and if the weather does not permit it, the team will test other features indoors. Table 12 in Appendix D lists the resources, access dates, and lead times for each milestone.

5.2 Materials and other resources

Table 7: Preliminary cost estimate by subsystem (CAD).

Subsystem	Items	Subtotal
Drive and power	Used motorized golf cart, motor controller, battery/power supply	375 CAD
Following and sensing	Raspberry Pi 5 and ultrasonic sensors (existing), lidar sensor	50 CAD
Club tracking	RFID read tags	20 CAD
Integration hardware	Mounting hardware, wiring, connectors, and switches	60 CAD
Other	Tax, shipping, and contingency	90 CAD
Estimated total		595 CAD

Table 7 groups the estimate by subsystem, and Table 13 in Appendix E itemizes it. The project will target a budget of approximately 600 CAD, based on estimated component prices and a used motorized golf cart sourced through Facebook Marketplace. The team already has a Raspberry Pi 5 and ultrasonic sensors, while university laboratory, fabrication, and software resources will be used as needed.

6. Conclusion

The proposed ASCC is a phone-free, autonomous golf pushcart that combines equipment transport with club tracking and distance information. Unlike existing follow carts, which only carry the clubs, the ASCC will use RFID-based club identification to keep an inventory of the bag and warn of a club left behind, and GPS to show the distance to the green. It will follow the golfer using sensor-based tracking, obstacle detection, and a remote emergency stop. The project will integrate these subsystems into a single working prototype, with testing progressing from controlled subsystem tests to outdoor validation with a golfer and golf bag. Seven milestones will guide development through final system testing by March 2027. An existing Raspberry Pi and ultrasonic sensors will be combined with purchased components and a used motorized golf cart, for an estimated total cost of 595 CAD. The completed prototype is expected to reduce the physical effort of managing golf equipment and to give the golfer club and distance information without a phone or a separate tracking device.

7. References

[1] Stewart Golf. “Q Follow electric golf trolley,” Accessed: Jul. 15, 2025. [Online]. Available: https://www.stewartgolf.co.uk/products/stewart-golf-q-follow

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[4] T. E. D. Hall, A. G. B. Willmott, N. J. Hynes, J. E. T. Wells, and P. M. Allen, “The validity and reliability of two golf performance-tracking Global Positioning System (GPS) devices,” *International Journal of Golf Science*, vol. 13, no. 1, pp. 103–121, 2025. [Online]. Available: https://www.golfsciencejournal.org/article/144587

[5] M. J. Islam, J. Hong, and J. Sattar, “Person-following by autonomous robots: A categorical overview,” *The International Journal of Robotics Research*, vol. 38, no. 14, pp. 1581–1618, Dec. 2019. DOI: 10.1177/0278364919881683

A. Detailed Comparison and Deliverables Tables

Tables 8 and 9 expand the summary tables in Section 1.

Table 8: Comparison of the proposed ASCC design with existing following-cart and shot-tracking products.

Design consideration	Stewart Golf Q Follow [1]	Arccos Smart Sensors with Apple Watch [2], [3]	Proposed ASCC design
Equipment transport	Motorized cart carries the golf bag	Separate bag-transport solution is required	Motorized cart carries the bag and the golf-assistance hardware
Golfer following	Follows a carried Bluetooth handset; also supports remote control	No powered following hardware	Tracks one golfer and maintains a specified following distance
Club detection	None	Club-mounted sensors identify the club used on detected shots	Bag-mounted reader detects tagged-club removal and return, and warns of a missing club
Distance information	Requires separate equipment	Watch shows GPS distances and recommends clubs	Cart display shows GPS distance to the green
Golfer input during play	Handset	Watch or phone	None, apart from a remote stop button

Table 9: Project deliverables, operating assumptions, and limitations.

Deliverable	Included functionality	Operating assumptions / constraints	Limitations / exclusions
Following-cart subsystem	Powered transport, one-golfer following, obstacle stopping, remote safety stop	Dry grass; defined terrain and speed limits; golfer detectable by the tracking system	No independent course navigation; restricted terrain capability
Smart-bag subsystem	Tagged-club identification, removal/return detection, club inventory, missing-club warning	One bag of tagged clubs; reliable reads within the defined bag configuration	Does not detect whether a removed club was used for a shot
Distance display	GPS distance to the green on a display screen, with no golfer input	Usable GPS; green coordinates loaded beforehand for the test site	Measured from the cart, not the ball; subject to GPS error

Deliverable	Included functionality	Operating assumptions / constraints	Limitations / exclusions
Integrated powered prototype	One working cart demonstrated with all three core subsystems	Battery sized for the specified demonstration; dry-weather testing	Waterproofing and full 18-hole endurance excluded
Extensions (not committed)	Shot-distance logging from club events, player club history and club recommendation, voice or button interaction	Attempted only after the core deliverables are verified	Not part of the success criteria

B. Detailed Safety Hazards and Project Risks

Figure 6 is the bow-tie diagram for loss of control of the cart, and Table 10 expands the summary in Table 4.

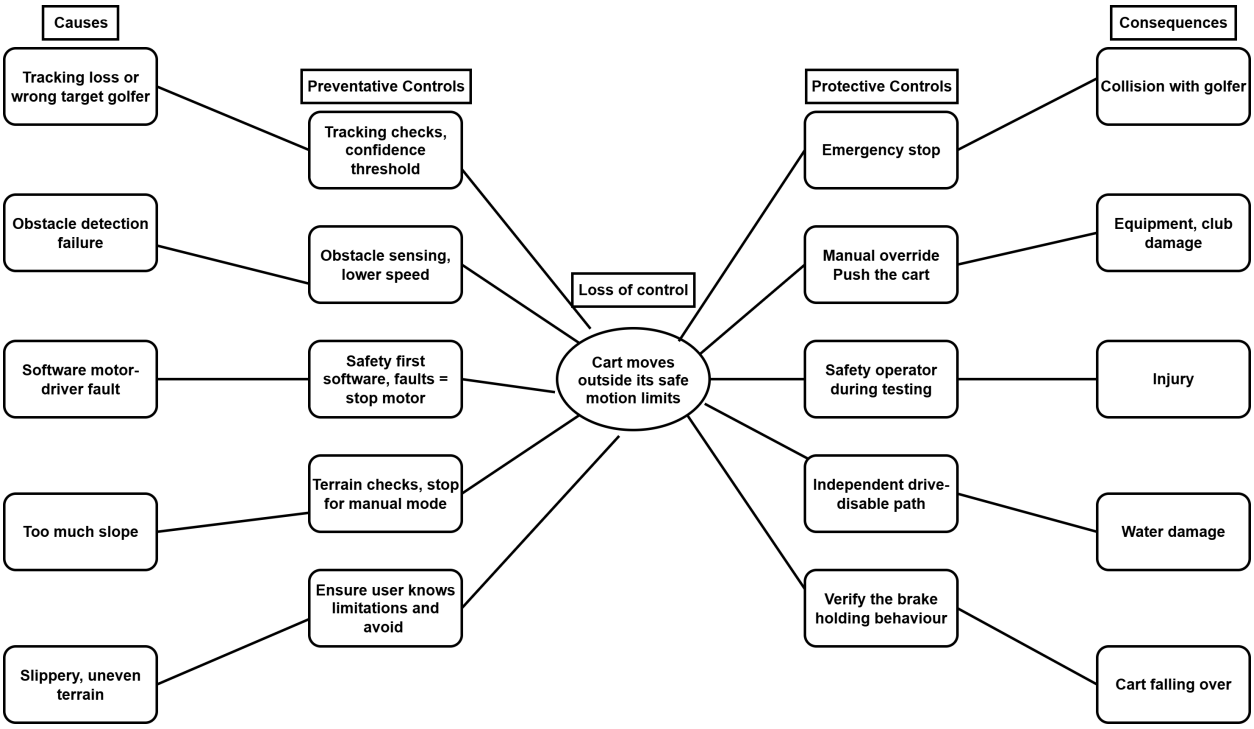


Figure 6: Bow-tie diagram for loss of control of the cart, showing causes and preventative controls on the left and protective controls and consequences on the right.

Table 10: Safety hazards and project risks with proposed mitigations.

Type	Hazard / risk	Possible consequence	Proposed mitigation
Safety	Tracking error or obstacle missed	Cart collides with a person or equipment	Limit speed; stop on tracking loss; obstacle detection; accessible emergency stop
Safety	Cart rolls or tips on a slope	Injury or equipment damage	Restrict terrain and payload; use a stable bag mounting; verify stopping and holding behaviour
Safety	Contact with wheels or drive mechanisms	Pinching or entanglement	Guard accessible moving parts; disable drive during adjustments
Safety	Battery fault, short circuit, or motor overload	Overheating, burns, or fire	Suitable battery protection, fuse, wiring, charger, and current limits
Safety	Water reaches electronics	Electrical faults or unintended behaviour	Dry-weather operation; protected electronics; stop testing if conditions deteriorate
Project	Club tags cannot be read reliably inside the bag	Club inventory fails to meet its accuracy target	Test the reader/tag arrangement early; simplify bag configuration or change sensing method
Project	GPS accuracy is insufficient	Incorrect green-distance information	Compare with reference distances; indicate unavailable data; define an achievable accuracy target
Project	Subsystems take too long to integrate	Incomplete MVP demonstration	Agree on interfaces early; integrate incrementally; keep red features optional

C. Detailed Test Plan

Table 11 expands the summary in Table 6.

Table 11: Proposed tests and the specifications each one verifies.

Requirement	Proposed test	Expected evidence of success
IF-01, IF-04, PF-01, PF-02, PF-03, PF-05	**Mobility and safety.** On a controlled grass area, the cart follows a walking golfer through straight lines and turns, meets soft obstacles, loses tracking, and receives a remote emergency stop.	Following-distance error, tracking range, and speed stay within their specified tolerances. The cart stops for every obstacle, tracking loss, and stop command, with reaction time, stop latency, and stop-link range recorded.
IF-03	**Club identification and presence.** With the bag stationary, clubs are removed and returned one at a time and several at once.	Every bag slot reports its club as present or absent correctly. Detection delay and missed and false events are recorded.
IF-02, IF-05	**GPS and onboard display.** At outdoor reference locations with predefined green coordinates, the displayed distance is compared with the reference distance.	Distance error stays within the specified accuracy, and the display shows the correct, readable information at each location.
PF-04	**Range.** The cart carries a loaded bag over an extended run on one charge.	The cart covers the specified distance without recharging.

D. Resource Access by Milestone

Table 12 connects the milestones in Table 5 to the resources each one needs.

Table 12: Resource access by milestone.

Milestone	Resource / source	Access date / lead time
M1	Motors, Controller, Golf Cart	Nov 1st, 2026; ASAP
M2	Raspberry Pi, Ultrasonic/Lidar Sensors	Nov 1st, 2026; ASAP
M3	RFID readers/tags, User Interface Display	Dec 1st, 2026; 2-3 weeks in advance
M4	Outdoor testing area	Feb 1st, 2027; no booking needed

E. Detailed Cost Estimate

Table 13 itemizes the subsystem totals in Table 7.

Table 13: Preliminary cost estimate (CAD).

Item / source	Qty.	Unit cost	Total
Raspberry Pi 5 - 4G	1	Existing	0 CAD
Ultrasonic Sensors	2	Existing	0 CAD
RFID Read Tags	2	10 CAD (5)	20 CAD
Lidar Sensor	1	50 CAD	50 CAD
Used Motorized Golf Cart	1	250 CAD	250 CAD
Motor controller	1	50 CAD	50 CAD
Battery/Power Supply	1	75 CAD	75 CAD
Mounting Hardware	1	30 CAD	30 CAD
Wiring, Connectors, and Switches	1	30 CAD	30 CAD
Tax and Shipping	-	-	75 CAD
Contingency	-	-	15 CAD
Estimated total			595 CAD